

Extending IS-MCTS for Secret Hitler with Neural Enhancements

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1 Introduction

Secret Hitler is a hidden-role, social deduction game involving incomplete information, probabilistic reasoning, and deception. Unlike perfect-information games such as Go or Chess, its mechanics demand reasoning under uncertainty and multi-agent interaction. This project builds on my previous work for CPSC 474, where I implemented an Information Set Monte Carlo Tree Search (IS-MCTS) agent for a simplified version of Secret Hitler in C++. That implementation compared favorably against random and greedy baselines but left two major avenues unexplored: full game complexity and more sophisticated state representation.

For my senior project, I propose to (i) restore full gameplay complexity, (ii) move from the single-observer IS-MCTS to a multi-observer variant, and (iii) explore augmenting IS-MCTS with a compact Transformer encoder to better model game state and inform search. Together, these directions will both deepen the algorithmic foundation of the project and test the feasibility of neural-assisted planning in a hidden-information, multi-agent setting.

2 Background

2.1 Secret Hitler

Secret Hitler is a hidden-role, social deduction game for 5–10 players. Each player is secretly assigned a role: most are Liberals, a minority are Fascists, and one Fascist is designated as Hitler. The Liberals win by enacting five Liberal policies or assassinating Hitler, while the Fascists win by enacting six Fascist policies or electing Hitler as Chancellor after three Fascist policies have passed.

Each round consists of a government election and a legislative session. The President nominates a Chancellor, and all players vote on the government; if the vote fails three times in a row, the top card of the policy deck is enacted automatically. If the vote succeeds, the President draws three policy cards, discards one, and passes two to the Chancellor, who must enact one. Because discards are hidden, players often make claims about what they saw, creating opportunities for deception and bluffing.

As Fascist policies accumulate, special powers such as investigations, deck peeks, or executions are unlocked, raising the strategic stakes. Success requires coalition building, reading opponents' intentions, and managing uncertainty, making Secret Hitler a rich testbed for studying reasoning and decision-making under imperfect information.

2.2 Information Set Monte Carlo Tree Search (IS-MCTS)

Monte Carlo Tree Search (MCTS) is a simulation-based planning algorithm that balances exploration and exploitation by repeatedly simulating possible futures and using the outcomes to guide decision-making. MCTS has been highly successful in perfect-information games like Go and Hex.

However, in imperfect-information games like Secret Hitler, standard MCTS fails because players cannot fully observe the game state. Information Set MCTS (IS-MCTS) addresses this by sampling possible complete states (called *determinizations*) consistent with a player's information. Each simulation proceeds as if the game were perfect information, and statistics are aggregated across many such determinizations. The single-observer version assumes only the current player reasons with hidden information, while the multi-observer

version allows all simulated agents to maintain and act according to their own beliefs, providing richer and more realistic strategies.

2.3 Transformers as State Encoders

Transformers are a neural network architecture originally introduced for sequence modeling in natural language processing. Unlike recurrent networks (e.g., LSTMs), which process inputs step-by-step, Transformers rely on *self-attention* mechanisms that allow each element in a sequence to attend to all others in parallel. This makes them effective at modeling long-range dependencies, relationships, and interactions.

In game AI, Transformers can represent not just linear sequences but also structured logs of events and relational data between players. For Secret Hitler, a Transformer could ingest the sequence of elections, votes, claims, and powers used, along with per-player embeddings. Through self-attention, it can learn patterns such as which players consistently vote together or contradict their past claims. When combined with IS-MCTS, a Transformer can serve as a learned function approximator, outputting policy priors and value estimates that guide the search more effectively.

2.4 Neural Encoders in Search Frameworks

AlphaZero demonstrated the power of combining deep networks with MCTS: a network predicts policy priors and state values, which are used to guide exploration and backpropagation. While AlphaZero focused on perfect-information, deterministic games, recent research (e.g., MCTransformer, neural IS-MCTS extensions) suggests that compact neural models can also benefit hidden-information settings by providing better state abstraction and action priors.

3 Project Description

The project has two main thrusts:

3.1 Multi-Observer IS-MCTS

Goal: Implement and evaluate a multi-observer IS-MCTS agent for Secret Hitler.

Tasks:

- Reconstruct prior codebase from course archives.
- Extend to multi-observer: each simulated agent maintains and updates its own belief distribution.
- Benchmark against the single-observer version and heuristic agents under full rules (5–7 players, powers enabled).

Expected Outcome: Demonstrate the effect of multi-observer reasoning on agent performance and realism of play.

3.2 Transformer-Based State Encoder (Exploratory)

Goal: Replace handcrafted feature extraction with a neural encoder that consumes game history and outputs policy/value estimates.

Architecture:

- Input tokens: sequence of past actions (elections, votes, claims, policies), per-player feature tokens, and private role embedding.
- Encoder: 2–3 layer Transformer ($d_{model} \approx 192$) with attention across events and players.
- Heads: (i) policy priors for legal actions, (ii) state value (team win probability).

Integration with IS-MCTS: Use policy priors to bias exploration (PUCT) and value head for leaf evaluation.

Expected Outcome: Even if not trained to full AlphaZero loop, the encoder should demonstrate feasibility of learned representations improving search guidance.

3.3 Evaluation

Baselines: Random agent, greedy agent, single-observer IS-MCTS.

Metrics: Win rate, computational efficiency (simulations per second), and quality of play (measured via game statistics such as election tracker usage, policy distribution, and consistency of strategies).

Stretch Goal: Human playtesting to evaluate realism of bot strategies.

4 Timeline

- Weeks 1–4: Implement multi-observer IS-MCTS. Benchmark against single-observer and heuristics under full rules.
- Weeks 5–8: Develop Transformer encoder prototype. Integrate into IS-MCTS (policy priors + value head). Train/test on self-play data.
- Weeks 9–10: Final experiments: compare pure IS-MCTS vs. neural-augmented search. Analyze results and prepare deliverables.

5 Deliverables

1. Codebase: C++ IS-MCTS engine with multi-observer support, plus Python/PyTorch modules for neural encoder.
2. Evaluation Report: Quantitative results comparing agent variants.
3. Poster & Presentation: Summary of methods, findings, and implications.
4. Final Report: A ~15-page paper detailing methodology, experiments, and analysis.

6 References

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